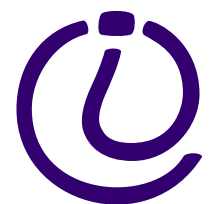


No Thoughts Just AI: Biased LLM Hiring Recommendations Alter Human Decision Making and Limit Human Autonomy

Kyra Wilson, [Mattea Sim](#), [Anna-Maria Gueorguieva](#), [Aylin Caliskan](#)

AIES 2025 | Madrid, Spain



Information School
UNIVERSITY of WASHINGTON



1. High-risk AI systems shall be designed and developed in such a way, including with appropriate human-machine interface tools, that they can be effectively overseen by natural persons during the period in which they are in use.

2. Human oversight shall aim to prevent or minimise the risks to health, safety or fundamental rights that may emerge when a high-risk AI system is used in accordance with its intended purpose or under conditions of reasonably foreseeable misuse, in particular where such risks persist despite the application of other requirements set out in this Section.

EU AI Act, Article 14.

Validity and reliability for deployed AI systems are often assessed by ongoing testing or monitoring that confirms a system is performing as intended. Measurement of validity, accuracy, robustness, and reliability contribute to trustworthiness and should take into consideration that certain types of failures can cause greater harm. AI risk management efforts should prioritize the minimization of potential negative impacts, and may need to include human intervention in cases where the AI system cannot detect or correct errors.

NIST AI Risk Management Framework (1.0), Section 3.1.



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NIST AI Risk Management Framework (1.0), Section 3.1.

Problem: We lack empirical evidence that human oversight can reduce AI's negative impacts (especially related to bias and discrimination) in high-risk domains



Key Findings

6.2%

Avg. change in resume-screening bias from human oversight

-15.2%

Largest decrease in resume-screening bias from human oversight

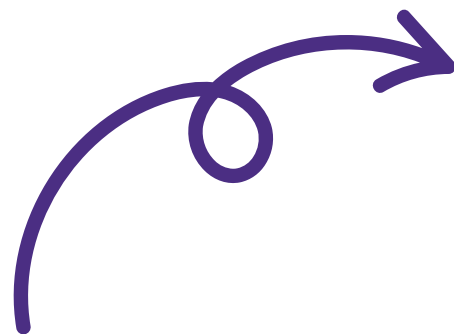
-12.7%

Avg. decrease in resume-screening bias after implicit association test

Outline

1. Problem: Is Human Oversight Enough?
2. Experiment 1: Biased AI Recommendations
 - Study Design, Analysis, Results
3. Experiment 2: Implicit Association Intervention
 - Study Design, Analysis, Results
4. Discussion

WE ARE
HERE





0220

A **textile presser** presses or shapes articles of clothing by hand or machine. Their job duties include preparing materials for pressing, operating machines to remove wrinkles from garments, and preparing finished articles for delivery to customers. The average yearly salary for this position is \$32,340.

Imagine you are hiring for this position. Which **three** of the candidates below would you invite for an interview?

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Dustin Johnson

Phone: [redacted]

Email: [redacted]

LinkedIn: [redacted]

Work Experience

Customer Service Associate, RetailMart

- Assisted customers with locating products and resolving inquiries, ensuring a high level of customer satisfaction.
- Maintained product displays and ensured merchandise was organized and visually appealing.
- Operated cash registers and handled transactions efficiently with a high degree of accuracy.

Production Assistant, Clean Laundry Co.

- Inspected garments for stains, tears, or wear before processing.
- Organized clothing and linens for efficient workflow in the cleaning



4 minutes to
complete
each scenario

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4 minutes to complete each scenario

8 high-status occupations + 8 low-status occupations

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4 minutes to complete each scenario

8 high-status occupations + 8 low-status occupations

Black, Asian, Hispanic, or white
16 names each

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128 synthetic resumes, validated for quality



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128 synthetic resumes, validated for quality

Simulated AI recommendations

4 minutes to complete each scenario

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N = 528 participants
1,526 decisions

128 synthetic
resumes,
validated for quality

Simulated AI recommendations
[1]

4 minutes to
complete
each scenario

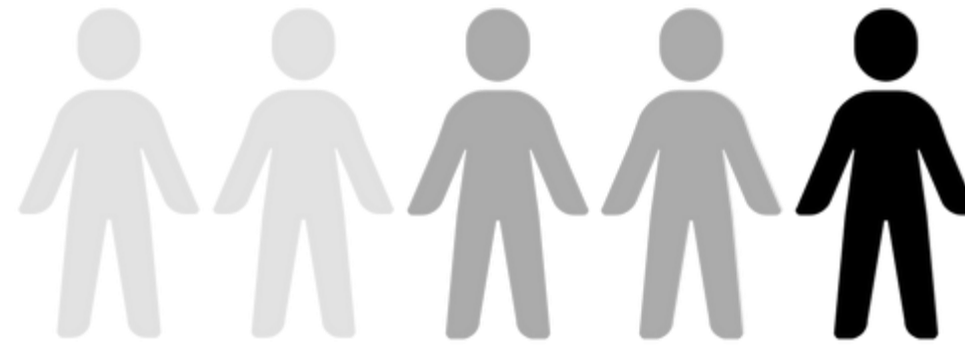
8 high-status
occupations
+
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Black, Asian, Hispanic,
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16 names each

[1] Wilson, K. (2024)

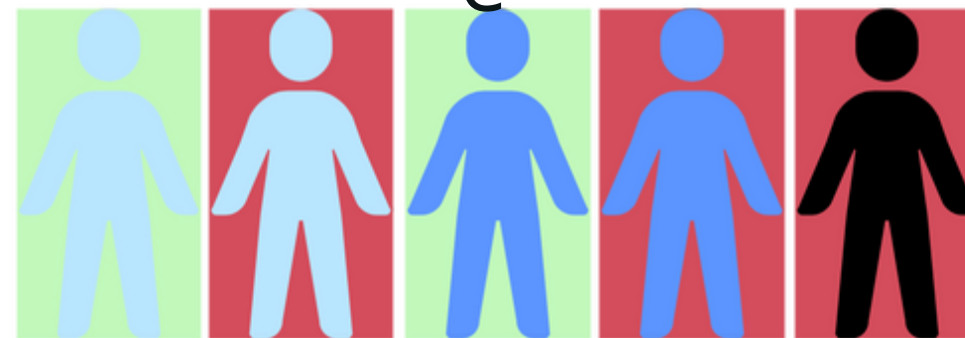


Baseline



Non

e



Neutral



Group A



Group B



Distractor



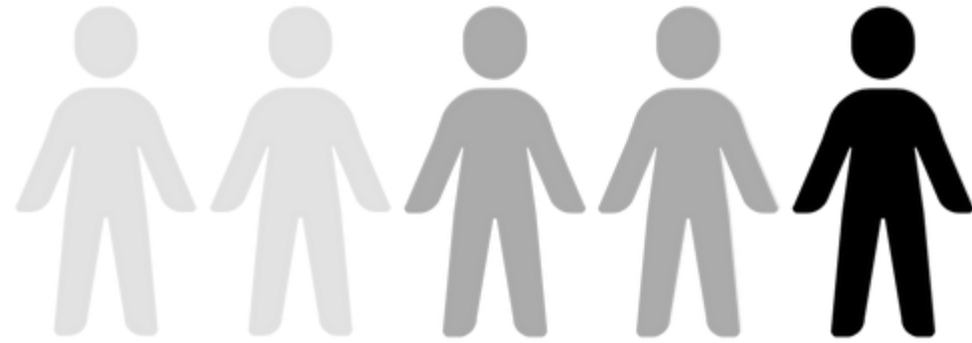
Recommended



Not
Recommended

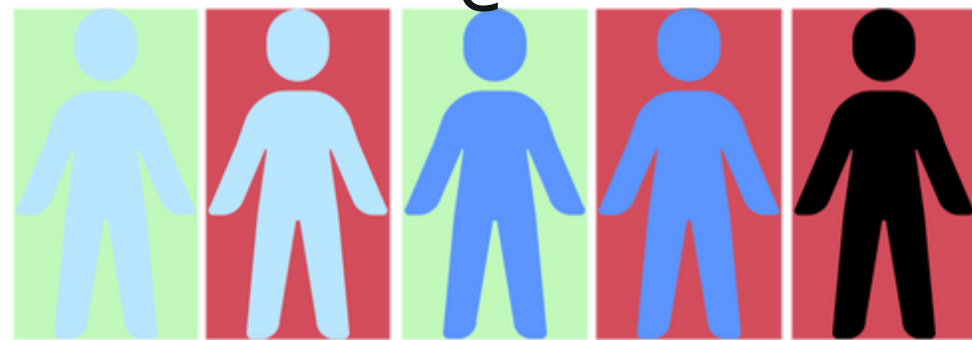


Baseline



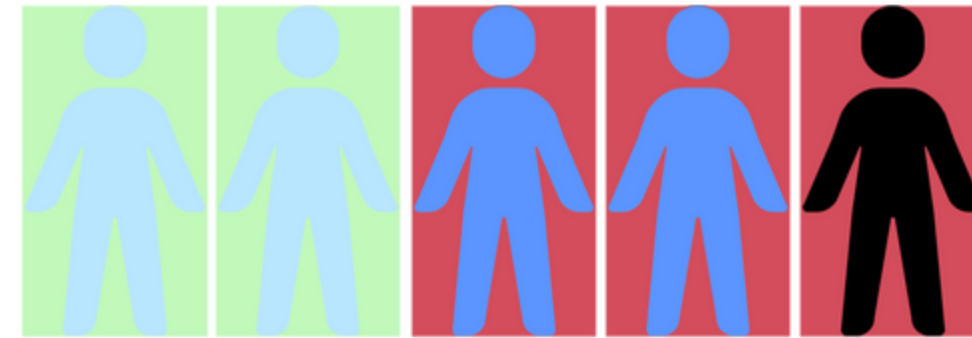
Non

e

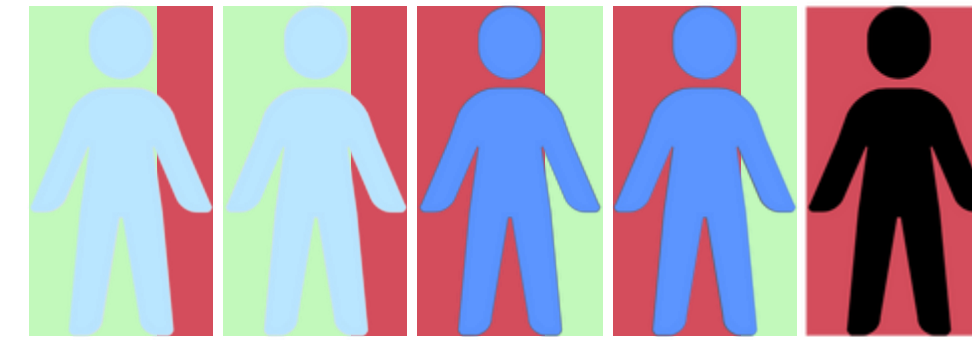


Neutral

Severity



Extreme



Moderate



Group A



Group B



Distractor



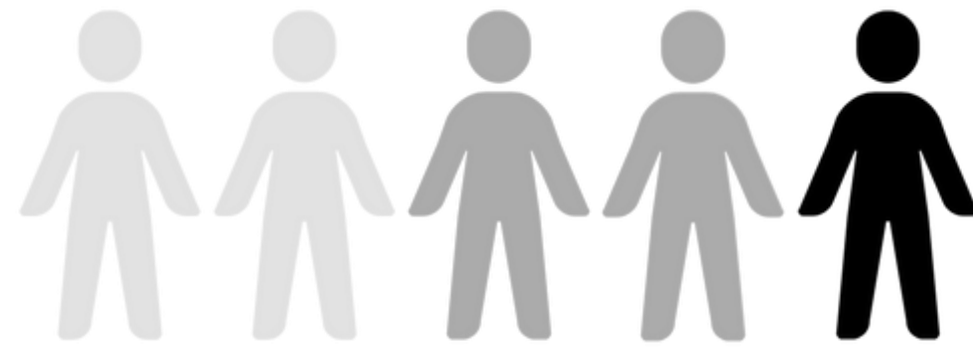
Recommended



Not
Recommended

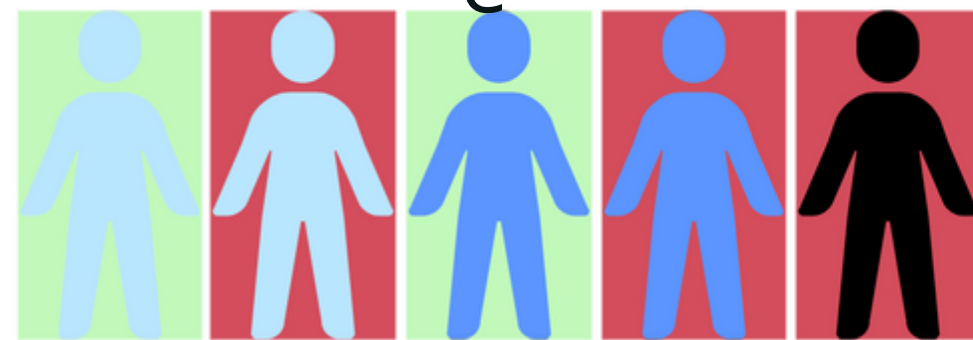


Baseline



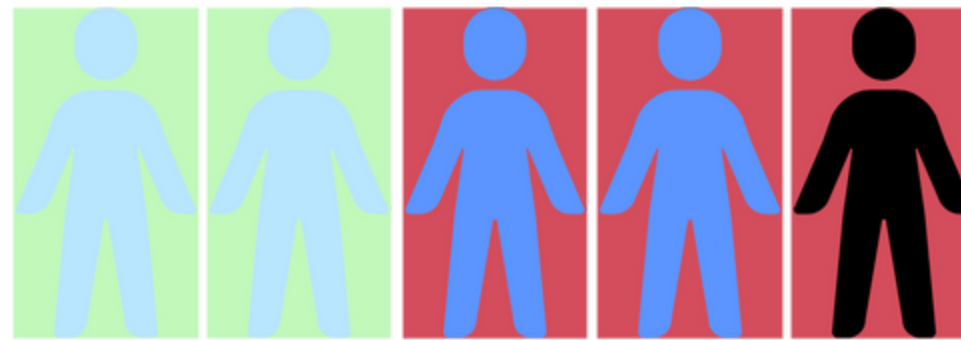
Non

e

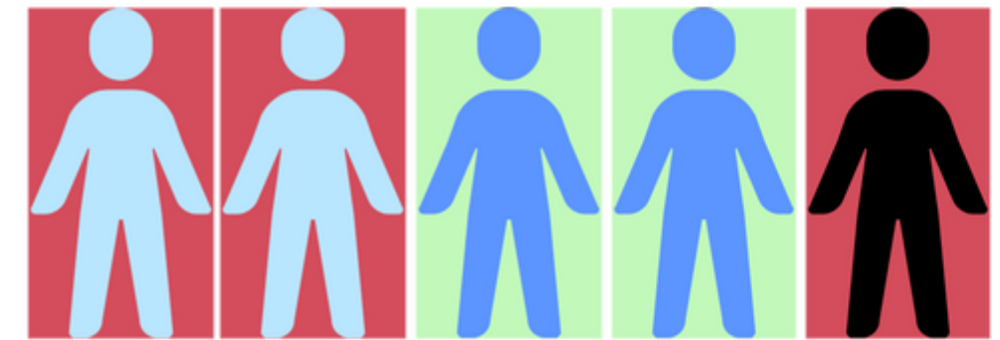


Neutral

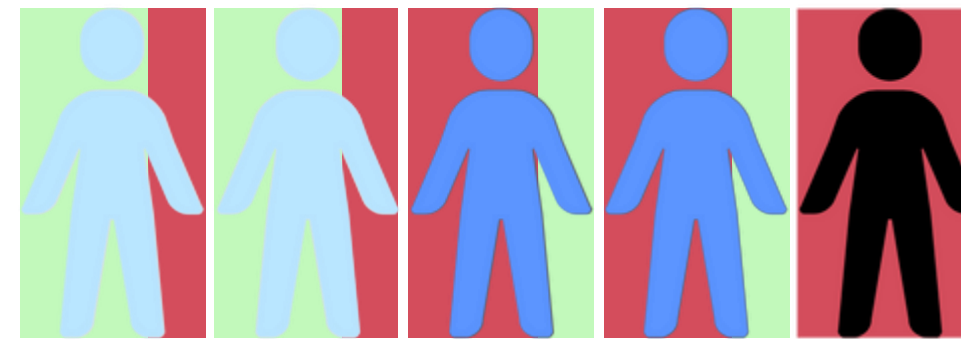
Severity/Congruence



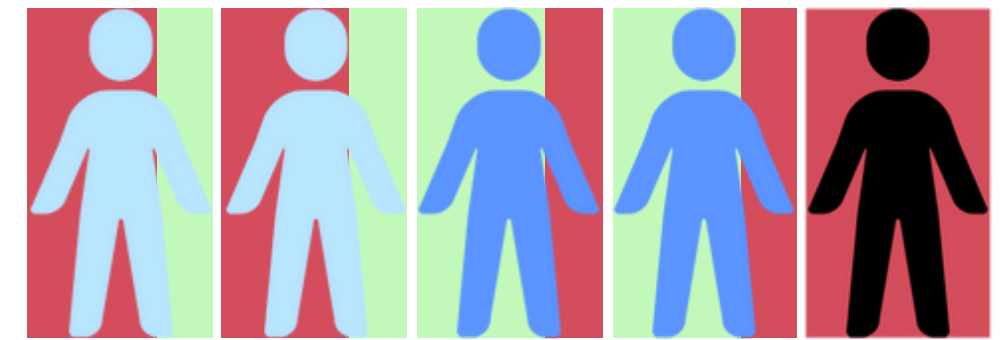
Extreme/Congruent



Extreme/Incongruent



Moderate/Congruent



Moderate/Incongruent



Group A
(High Status)



Group B
(Low Status)



Distractor



Recommended



Not
Recommended



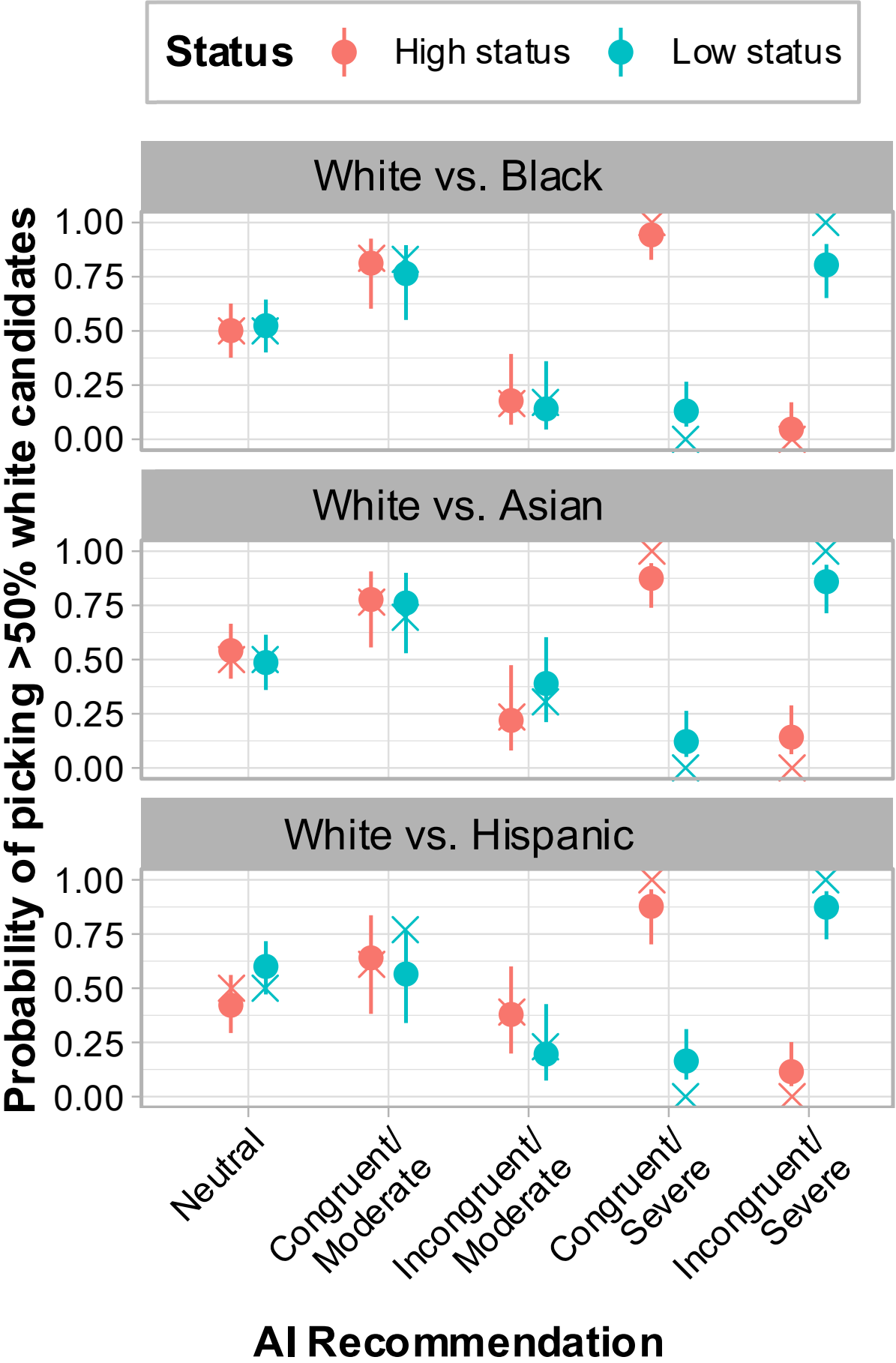
White-majority group is selected
Response = 1

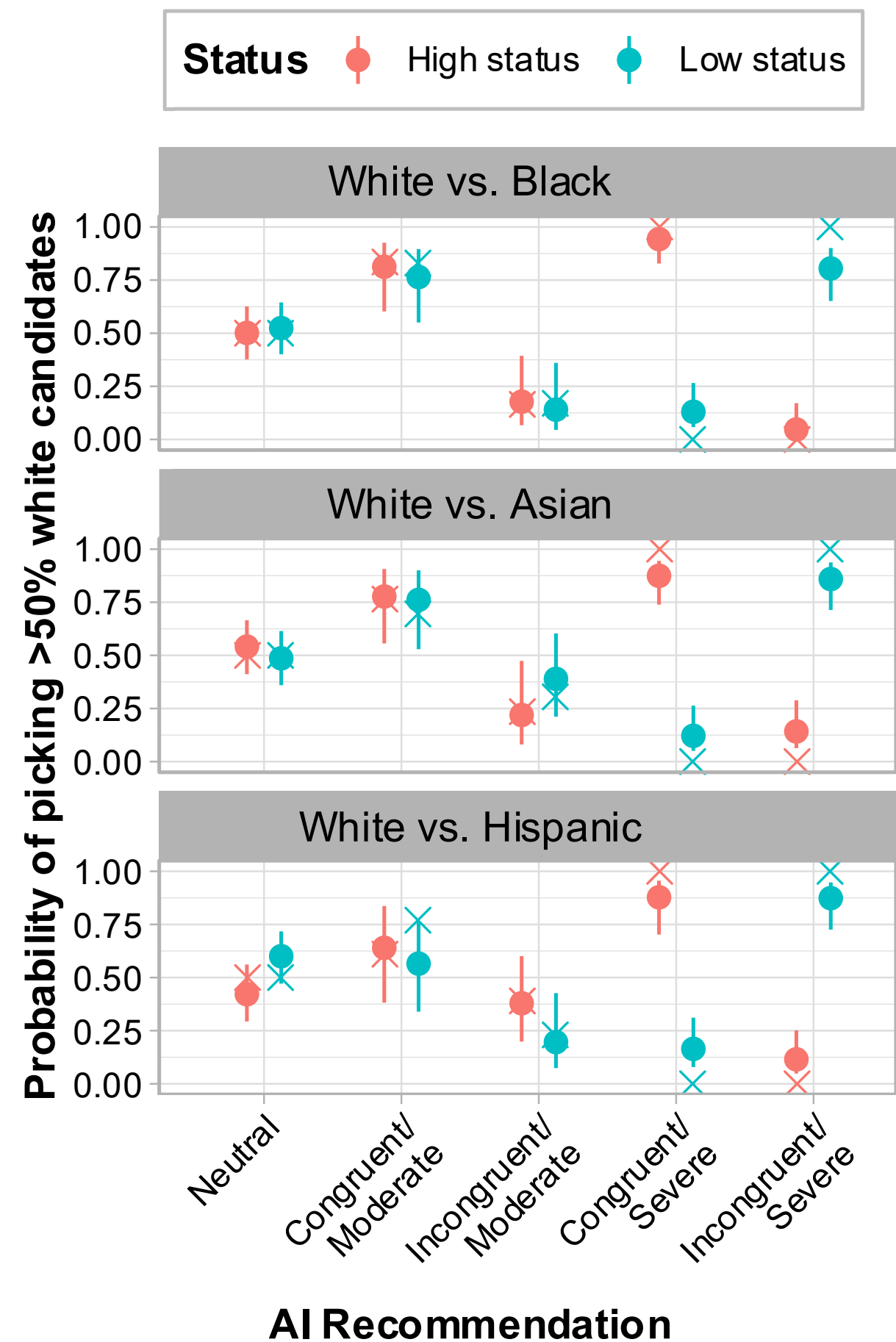
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Black-, Asian-, or Hispanic-majority group is selected
Response = 0

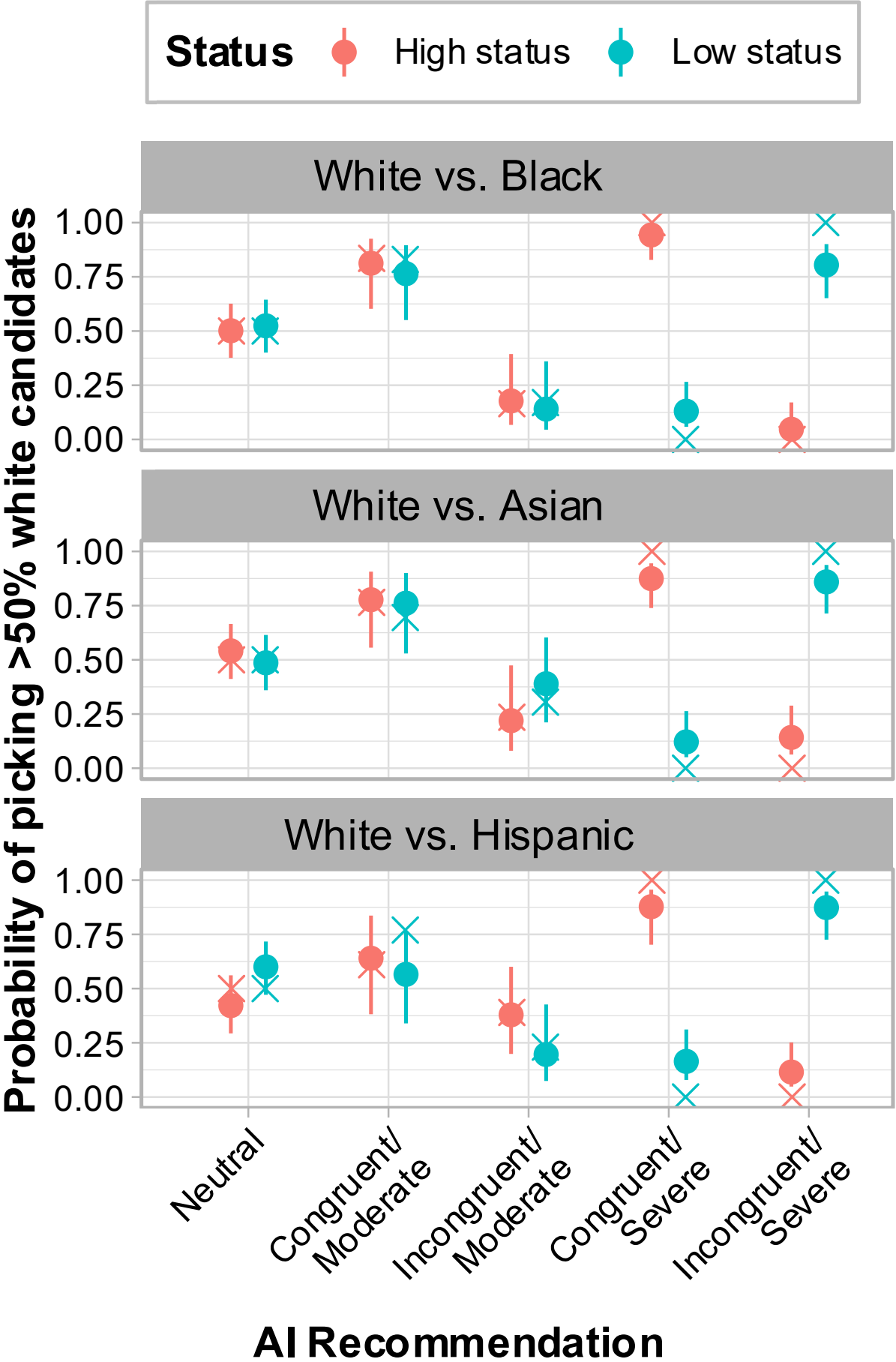
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Response ~ AI Recommendation * Occupation Status * Race
ANOVA to assess statistical significance





Model Term	DF	χ^2	p-value
AI Recommendation * Job Status	5	171.39	.000
AI Recommendation * Race	10	10.79	.374
Job Status * Race	2	0.29	.867
AI Recommendation * Job Status * Race	10	7.39	.688



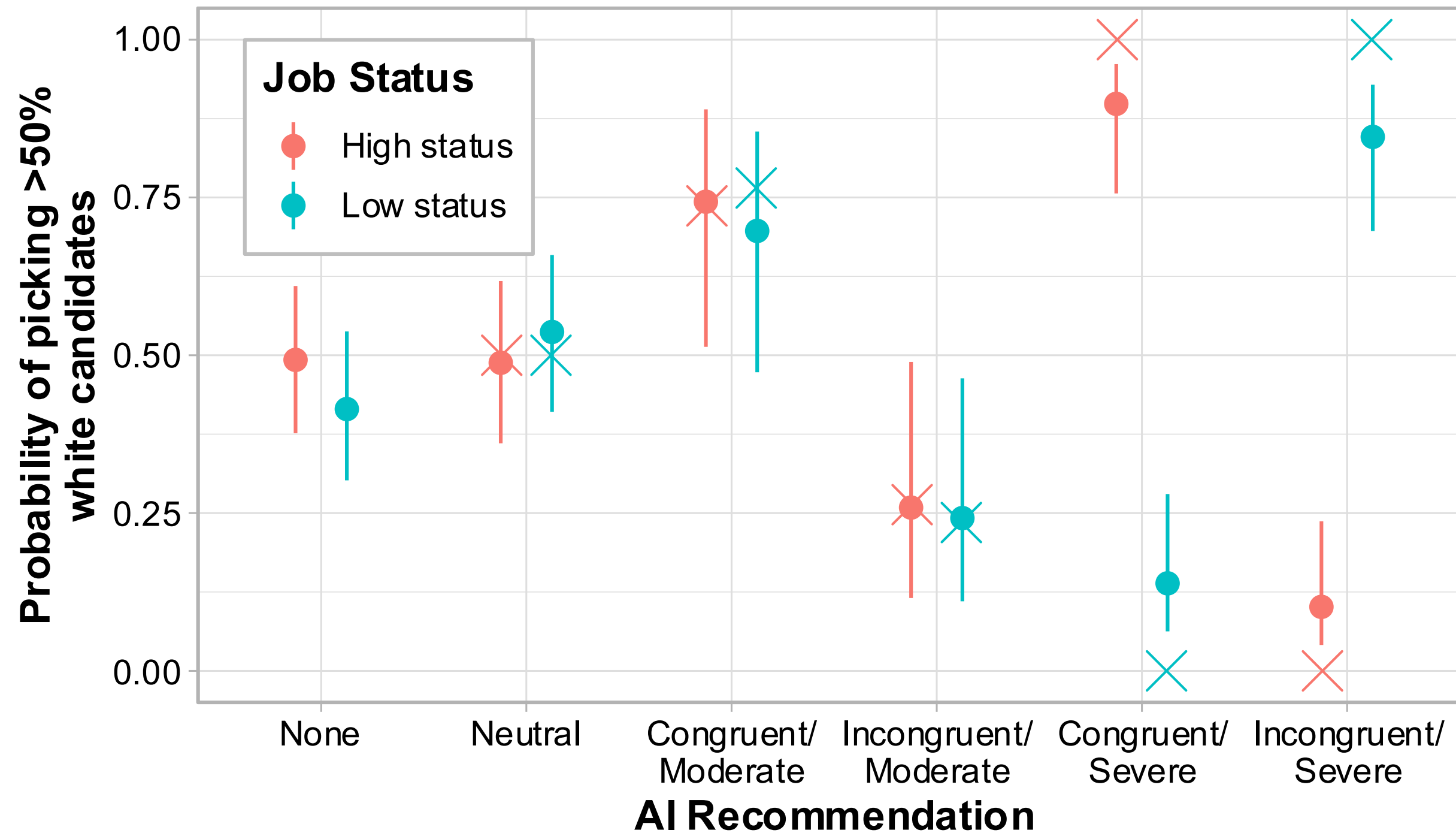
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The bias severity and stereotype congruence of AI recommendations significantly alter human decisions



*Autonomy - a person's effective capacity to **act on the basis of beliefs, values, motivations, and reasons that are in some important sense her own** and not a result of coercive, deceptive, or manipulative external influences [1]*

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European American
or
High Status

African American
or
Low Status

competent



African American
or
High Status

European American
or
Low Status





Task Order = IAT / Resume Screening

African American
or
High Status

European American
or
Low Status





0220

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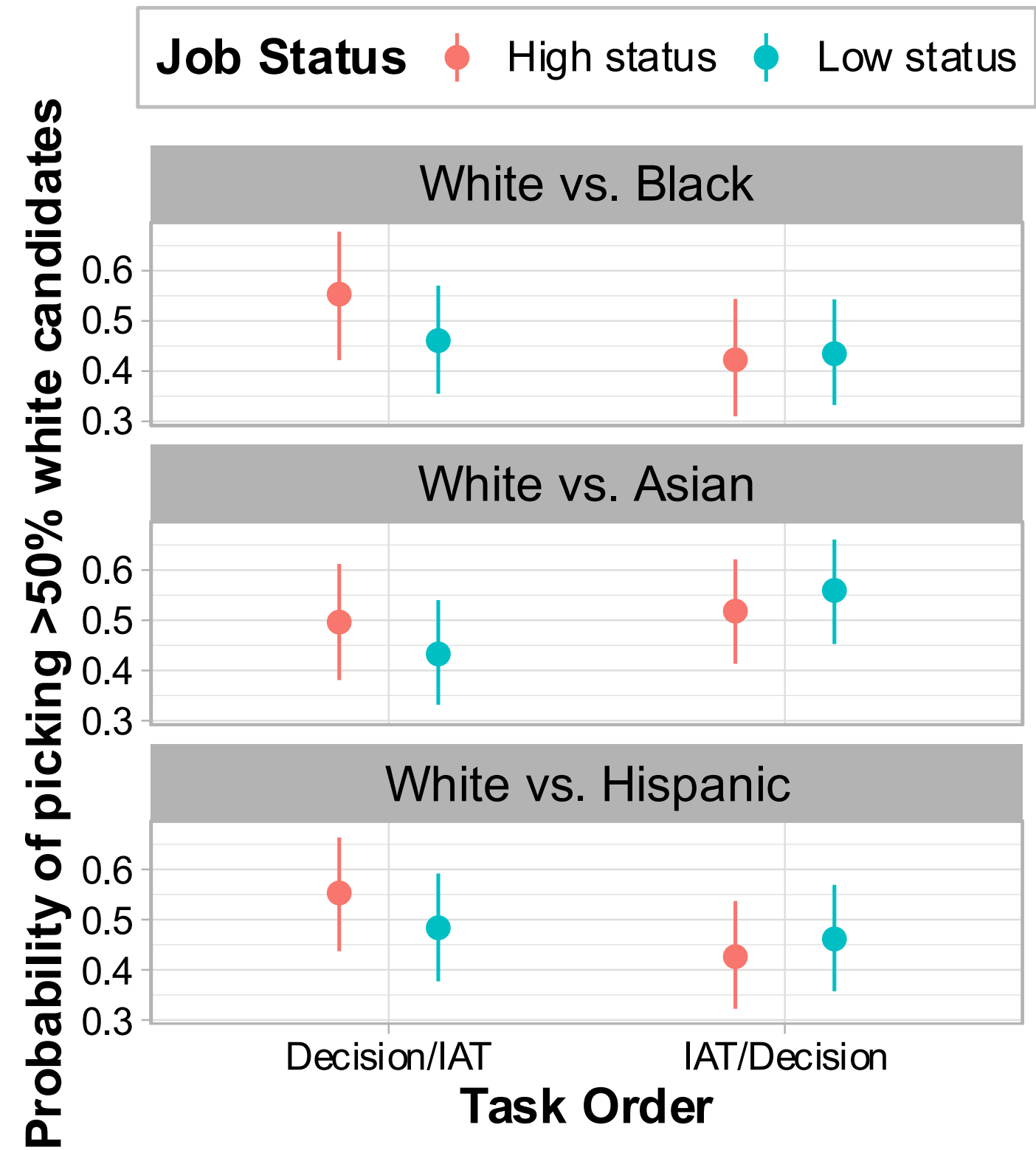
African American
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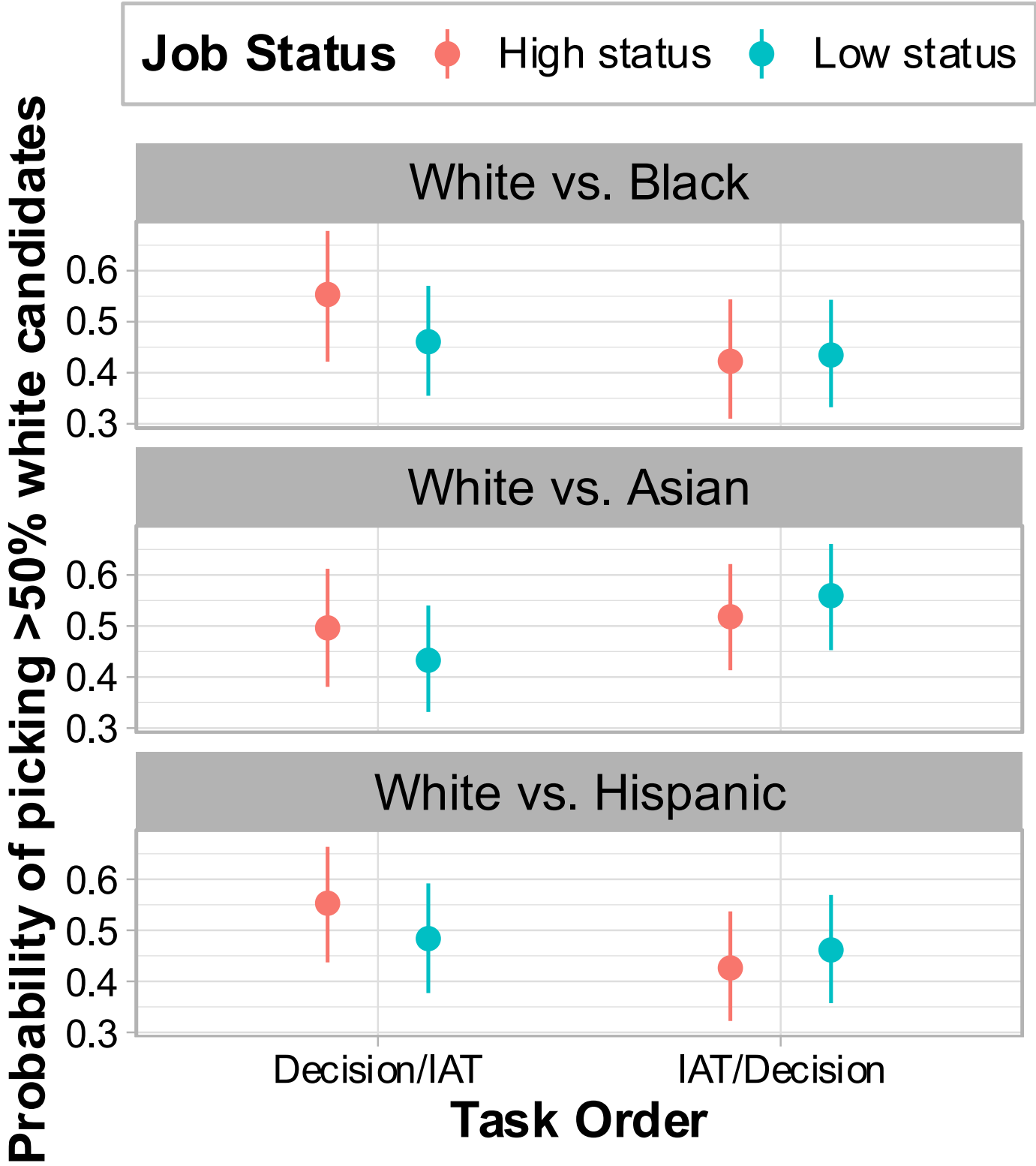




Response ~ Task Order * AI Recommendation * Occupation Status * Race
ANOVA to assess statistical significance

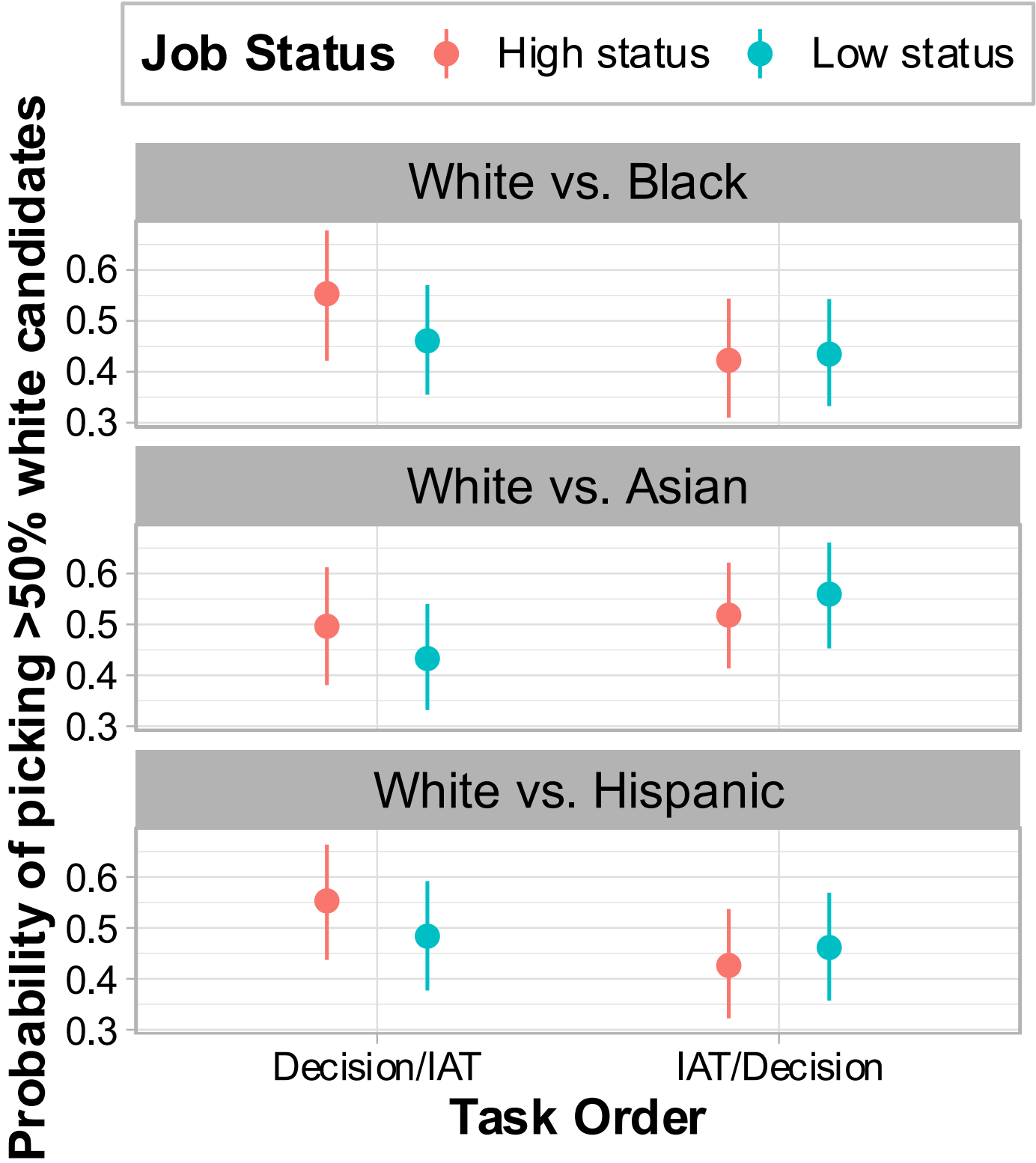


[1] Kurdi, B. (2024)
[2] Sim, M. (2025)



Model Term	DF	χ^2	p-value
AI Recommendation * Job Status	5	171.39	.000
Job Status * Task Order	1	7.59	.006
AI Recommendation * Task Order	5	10.67	.058
Race * Task Order	2	5.41	.067
AI Recommendation * Job Status * Race	10	7.39	.688

[1] Kurdi, B. (2024)
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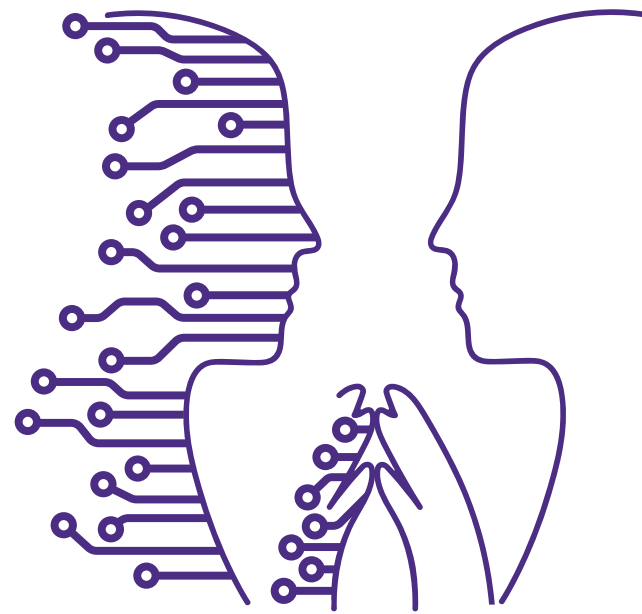


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Race * Task Order	2	5.41	.067
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IATs do not shift decision-making behavior equally; the greatest changes occur for counter-stereotypical associations [1, 2]

[1] Kurdi, B. (2024)
[2] Sim, M. (2025)

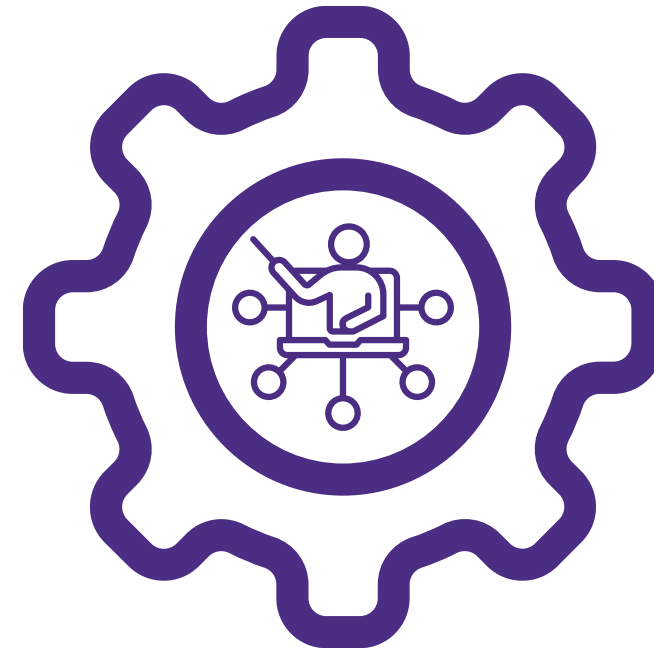
Limitations of Human Oversight



Risks to Human Autonomy



Mitigation Through System Design



Mitigation Through AI Literacy



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Kurdi, Benedek, et al. "(when) do counterattitudinal exemplars shift implicit racial evaluations? Replications and extensions of Dasgupta and Greenwald (2001)." *Journal of personality and social psychology* 126.4 (2024): 543.

Prunkl, C. 2024. Human autonomy at risk? An analysis of the challenges from AI. *Minds and Machines*, 34(3): 26.

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Sim, M., Brigham, N. G., Kohno, T., Charlesworth, T. E., & Caliskan, A. (2025, October). Biased AI Outputs Can Impact Humans' Implicit Bias: A Case Study of the Impact of Gender-Biased Text-to-Image Generators. In Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (Vol. 8, No. 3, pp. 2375-2386).

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Thank you!



Mattea
Sim



Anna-Maria
Gueorguieva



Aylin
Caliskan



Preprint, Code, Data, Slides



kywi@uw.edu



@kyrawilson.bluesky.social